PROPOSITIONAL LOGIC

LECTURE ONE

Course Objectives

- Introduce Mathematical Logic.
- Introduce Proof Techniques.
- Develop understanding of counting, functions and relations.

Course Outcome

At the end of this course the students will be able to:

- Distinguish between propositional logic and first order logic.
- Know how to check if a proposition is satisfiable.
- Apply induction and other proof techniques towards solving problems in elementary algebra.
- Understand elementary combinatorics.
- Distinguish between functions and relations.

Grading Policy

Assignment, tutorials, and homework	40%
Midterm Exam	20%
Final Exam	40%
Total	100%

References

- Discrete Mathematics and its Applications Paperback – August 10, 2018, by NA.
- Epp, S. (2010) Discrete Mathematics with applications. 4th Ed. Brooks: Cole.

Lecture Outlines

- Proposition
  - Statement, Truth value,
  - Proposition, Propositional symbol, Open proposition
- Operators
  - Define by truth tables
  - Composite propositions
  - Tautology and contradiction
- Equivalence of propositional statements
  - Definition
  - Proving equivalence (by truth table or equivalence laws)

Logic

- Important for program design
- Used for designing electronic circuitry

- (Propositional)Logic is a system based on propositions.
- A proposition is a (declarative) statement that is either true or false (not both).
- We say that the truth value of a proposition is either true (T) or false (F).
- Corresponds to 1 and 0 in digital circuits

The Statement/Proposition Game

- "Elephants are bigger than mice."

Is this a statement? yes

Is this a proposition? yes

What is the truth value of the proposition? true

The Statement/Proposition Game

- "520 < 111"

Is this a statement? yes

Is this a proposition? yes

What is the truth value of the proposition? false

The Statement/Proposition Game

- "y > 5"

Is this a statement? yes

Is this a proposition? no

Its truth value depends on the value of y, but this value is not specified.

We call this type of statement a propositional function or open sentence.

The Statement/Proposition Game

- "Today is May 27 and 99 < 5."

Is this a statement? yes

Is this a proposition? yes

What is the truth value of the proposition? false

The Statement/Proposition Game

- "Please do not fall asleep."

Is this a statement? **no**

It's a request.

Is this a proposition? **no**

Only statements can be propositions.

The Statement/Proposition Game

- "If the moon is made of cheese,
- then I will be rich."

Is this a statement? yes

Is this a proposition? yes

What is the truth value of the proposition? probably true

The Statement/Proposition Game

- "x < y if and only if y > x."

Is this a statement? yes

Is this a proposition? yes

… because its truth value
does not depend on
specific values of x and y.

What is the truth value
of the proposition? true

Combining Propositions

- As we have seen in the previous examples, one or more propositions can be combined to form a single compound proposition.

- We formalize this by denoting propositions with letters such as p, q, r, s, and introducing several logical operators or logical connectives.

Logical Operators (Connectives)

•We will examine the following logical operators:

- Negation (NOT, $\neg$)
- Conjunction (AND, $\wedge$)
- Disjunction (OR, $\vee$)
- Exclusive-or (XOR, $\oplus$)
- Implication (if – then, $\rightarrow$)
- Biconditional (if and only if, $\leftrightarrow$)

•Truth tables can be used to show how these operators can combine propositions to compound propositions.

Logical operators

Lecture two

Negation (NOT)

- Unary Operator, Symbol: ¬

P	¬ P
true (T)	false (F)
false (F)	true (T)

Conjunction (AND)

- Binary Operator, Symbol: $\wedge$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR)

- Binary Operator, Symbol: $\vee$

P	Q	P $\vee$ Q
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Or (XOR)

- Binary Operator, Symbol: $\oplus$

P	Q	P⊕Q
T	T	F
T	F	T
F	T	T
F	F	F

Implication (if - then)

- Binary Operator, Symbol: $\rightarrow$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional (if and only if)

- Binary Operator, Symbol: ↔

P	Q	P↔Q
T	T	T
T	F	F
F	T	F
F	F	T

Statements and Operators

- Statements and operators can be combined in any way to form new statements.

P	Q	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Statements and Operations

- Statements and operators can be combined in any way to form new statements.

P	Q	$P \wedge Q$	$\neg P$	$\neg Q$	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$
T	T	T	F	F	F	F
T	F	F	F	T	T	T
F	T	F	T	F	T	T
F	F	F	T	T	T	T

Exercises

- To take discrete mathematics, you must have taken calculus or a course in computer science.
- When you buy a new car from Acme Motor Company, you get $2000 back in cash or a 2% car loan.
- School is closed if more than 2 feet of snow falls or if the wind chill is below -100.

Exercises

- To take discrete mathematics, you must have taken calculus or a course in computer science.
  - P: take discrete mathematics
  - Q: take calculus
  - R: take a course in computer science
- $P \rightarrow \mathbf{Q} \vee R$
- Problem with proposition R.

Exercises

- When you buy a new car from Acme Motor Company, you get $2000 back in cash or a 2% car loan.
  - P: buy a car from Acme Motor Company
  - Q: get $2000 cash back
  - R: get a 2% car loan
- $P \rightarrow \mathbf{Q} \oplus R$
- Why use XOR here? – example of ambiguity of natural languages

Exercises

- School is closed if more than 2 feet of snow falls or if the wind chill is below -100.
  - P: School is closed
  - Q: 2 feet of snow falls
  - R: wind chill is below -100
- $Q \wedge R \rightarrow P$
- Precedence among operators:
- $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$

Equivalent Statements

P	Q	$(P \wedge Q)$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
T	T	T	F	F	F	F	T
T	F	F	T	F	T	T	T
F	T	F	T	T	F	T	T
F	F	F	T	T	T	T	T

- The statements $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are logically equivalent, since they have the same truth table, or put it in another way, $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$ is always true.

Tautologies and Contradictions

- A tautology is a statement that is always true.
- Examples:
  - $R \vee (\neg R)$
  - $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
- A contradiction is a statement that is always false.
- Examples:
  - $R \wedge (\neg R)$
  - $\neg(\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q))$
- The negation of any tautology is a contradiction, and the negation of any contradiction is a tautology.

Equivalence

- Definition: two propositional statements S1 and S2 are said to be (logically) equivalent, denoted $S1 \equiv S2$ if
  - They have the same truth table, or
  - $S1 \Leftrightarrow S2$ is a tautology
- Equivalence can be established by
  - Constructing truth tables
  - Using equivalence laws (Table 5 in Section 1.2)

Equivalence

- Equivalence laws
  - Identity laws, $P \wedge T \equiv P$,
  - Domination laws, $P \wedge F \equiv F$,
  - Idempotent laws, $P \wedge P \equiv P$,
  - Double negation law, $\neg (\neg P) \equiv P$
  - Commutative laws, $P \wedge Q \equiv Q \wedge P$,
  - Associative laws, $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$,
  - Distributive laws, $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$,
  - De Morgan's laws, $\neg (P \wedge Q) \equiv (\neg P) \vee (\neg Q)$
  - Law with implication $P \rightarrow Q \equiv \neg P \vee Q$

Exercises

- Show that $P \rightarrow Q \equiv \neg P \vee Q$: by truth table
- Show that $(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R)$: by equivalence laws (q20, p27):
  - Law with implication on both sides
  - Distribution law on LHS

QUESTIONS?

Propositional Functions And Quantifications

Lecture three

Outlines

Propositional Function.
Universal Quantification & Existential Quantification.
Negation.
Nested Quantifiers.

Propositional Functions

Propositional function (open sentence) is statement involving one or more variables.

e.g.: $x-3 > 5$.

Let us call this propositional function P(x),

where P is the predicate and x is the variable.

- What is the truth value of P(2) ? false
- What is the truth value of P(8) ? false
- What is the truth value of P(9) ? true

When a variable is given a value, it is said to be instantiated.

Truth value depends on value of variable.

Universal Quantification

Let P(x) be a predicate (propositional function).

Universally quantified sentence:
For all x in the universe of discourse P(x) is true.

Using the universal quantifier ∀:
∀x P(x) "for all x P(x)" or "for every x P(x)"

(Note: ∀x P(x) is either true or false, so it is a proposition, not a propositional function.)

Universal Quantification

Example: Let the universe of discourse be all people

$S(x)$: x is a ISU student.

$G(x)$: x is a genius.

What does $\forall x\,(S(x) \rightarrow G(x))$ mean ?

"If x is a ISU student, then x is a genius." or

"All ISU students are geniuses."

If the universe of discourse is all ISU students, then the same statement can be written as

$\forall x\, G(x)$

Existential Quantification

Existentially quantified sentence:

There exists an x in the universe of discourse for which P(x) is true.

Using the existential quantifier ∃:

∃x P(x) "There is an x such that P(x)."

"There is at least one x such that P(x)."

(Note: ∃x P(x) is either true or false, so it is a proposition, but no propositional function.)

Existential Quantification

Example:

$P(x)$: x is a ISU professor.

$G(x)$: x is a genius.

What does $\exists x\,(P(x) \wedge G(x))$ mean ?

"There is an x such that x is a ISU professor and x is a genius."

or

"At least one ISU professor is a genius."

Quantification

Another example:
Let the universe of discourse be the real numbers.

What does $\forall x \exists y\ (x + y = 320)$ mean ?

"For every x there exists a y so that $x + y = 320$."
Is it true? yes
Is it true for the natural numbers? no

Disproof by Counterexample

A counterexample to $\forall x\ P(x)$ is an object c so that P(c) is false.

Statements such as $\forall x\ (P(x) \rightarrow Q(x))$ can be disproved by simply providing a counterexample.

Statement: "All birds can fly."
Disproved by counterexample: Penguin.

Negation

$\neg(\forall x\, P(x))$ is logically equivalent to $\exists x\, (\neg\, P(x))$.

$\neg(\exists x\, P(x))$ is logically equivalent to $\forall x\, (\neg\, P(x))$.

This is de Morgan's law for quantifiers

Negation

Examples

Not all roses are red

$\neg\forall x\ (\text{Rose}(x) \rightarrow \text{Red}(x))$

$\exists x\ (\text{Rose}(x) \wedge \neg\text{Red}(x))$

Nobody is perfect

$\neg\exists x\ (\text{Person}(x) \wedge \text{Perfect}(x))$

$\forall x\ (\text{Person}(x) \rightarrow \neg\text{Perfect}(x))$

Nested Quantifier

A predicate can have more than one variables.

S(x, y, z): z is the sum of x and y

F(x, y): x and y are friends

We can quantify individual variables in different ways

$\forall x, y, z\ (S(x, y, z) \rightarrow (x <= z \wedge y <= z))$

$\exists x\ \forall y\ \forall z\ (F(x, y) \wedge F(x, z) \wedge (y != z) \rightarrow \neg F(y, z)$

Nested Quantifier

Exercise: translate the following English sentence into logical expression

> "There is a rational number in between every pair of distinct rational numbers"

Use predicate $Q(x)$, which is true when x is a rational number

$\forall x,y\ (Q(x) \wedge Q(y) \wedge (x < y) \rightarrow \exists u\ (Q(u) \wedge (x < u) \wedge (u < y)))$

Summary

Propositional functions (predicates)

Universal and existential quantifiers, and the duality of the two

When predicates become propositions

- All of its variables are instantiated
- All of its variables are quantified

Nested quantifiers

- Quantifiers with negation

Logical expressions formed by predicates, operators, and quantifiers

Set Theory

Lecture Four

Set Theory

Set: Collection of objects (called elements)

$a \in A$ "a is an element of A"
"a is a member of A"

$a \notin A$ "a is not an element of A"

$A = \{a_1, a_2, \ldots, a_n\}$ "A contains $a_1, \ldots, a_n$"

Order of elements is insignificant

It does not matter how often the same element is listed (repetition doesn't count).

Set Equality

- Sets A and B are equal if and only if they contain exactly the same elements.
- Examples:
- A = {9, 2, 7, -3}, B = {7, 9, -3, 2} : A = B
- A = {dog, cat, horse},
  B = {cat, horse, squirrel, dog} : A ≠ B
- A = {dog, cat, horse},
  B = {cat, horse, dog, dog} : A = B

Examples for Sets

- "Standard" Sets:
- Natural numbers $\mathbf{N} = \{0, 1, 2, 3, \ldots\}$
- Integers $\mathbf{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
- Positive Integers $\mathbf{Z^+} = \{1, 2, 3, 4, \ldots\}$
- Real Numbers $\mathbf{R} = \{47.3, -12, \pi, \ldots\}$
- Rational Numbers $\mathbf{Q} = \{1.5, 2.6, -3.8, 15, \ldots\}$
- (correct definitions will follow)

Examples for Sets

- $A = \emptyset$ "empty set/null set"
- $A = \{z\}$ Note: $z \in A$, but $z \neq \{z\}$
- $A = \{\{b, c\}, \{c, x, d\}\}$ set of sets
- $A = \{\{x, y\}\}$ Note: $\{x, y\} \in A$, but $\{x, y\} \neq \{\{x, y\}\}$
- $A = \{x \mid P(x)\}$ "set of all x such that P(x)"

➢ P(x) is the membership function of set A

➢ $\forall x\,(P(x) \rightarrow x \in A)$

➢ $A = \{x \mid x \in \mathbf{N} \wedge x > 7\} = \{8, 9, 10, \ldots\}$

Examples for Sets

- We are now able to define the set of rational numbers Q:
- $\mathbf{Q} = \{a/b \mid a \in \mathbf{Z} \wedge b \in \mathbf{Z}^{+}\}$, or
- $\mathbf{Q} = \{a/b \mid a \in \mathbf{Z} \wedge b \in \mathbf{Z} \wedge b \neq 0\}$

- And how about the set of real numbers R?

- $\mathbf{R} = \{r \mid r \text{ is a real number}\}$

That is the best we can do. It can neither be defined by enumeration nor builder function.

Subsets

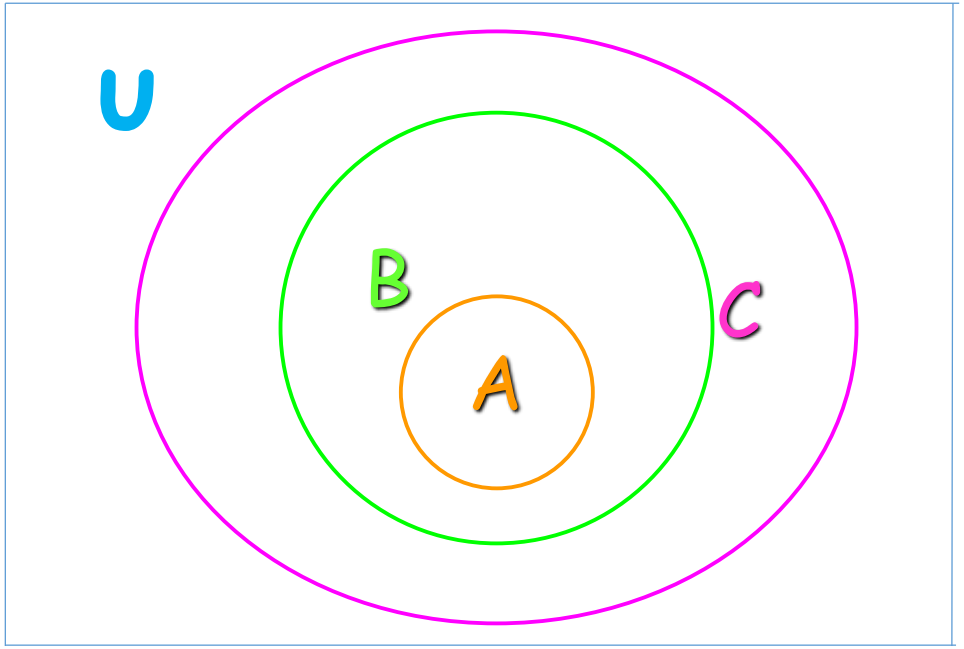
- $A \subseteq B$ "A is a subset of B"
- $A \subseteq B$ if and only if every element of A is also an element of B.
- We can completely formalize this:
- $A \subseteq B \Leftrightarrow \forall x\,(x \in A \rightarrow x \in B)$

- Examples:

$A = \{3, 9\}, B = \{5, 9, 1, 3\},$	$A \subseteq B$?	true
$A = \{3, 3, 3, 9\}, B = \{5, 9, 1, 3\},$	$A \subseteq B$?	true
$A = \{1, 2, 3\}, B = \{2, 3, 4\},$	$A \subseteq B$?	false

Subsets

- Useful rules:
- $A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$
- $(A \subseteq B) \wedge (B \subseteq C) \Rightarrow A \subseteq C$ (see Venn Diagram)



Subsets

- Useful rules:
- $\emptyset \subseteq A$ for any set A
- (but $\emptyset \in A$ may not hold for any set A)
- $A \subseteq A$ for any set A

- Proper subsets:
- $A \subset B$ "A is a proper subset of B"
- $A \subset B \Leftrightarrow \forall x\,(x \in A \rightarrow x \in B) \wedge \exists x\,(x \in B \wedge x \notin A)$
- or
- $A \subset B \Leftrightarrow \forall x\,(x \in A \rightarrow x \in B) \wedge \neg\forall x\,(x \in B \rightarrow x \in A)$

Cardinality of Sets

- If a set S contains n distinct elements, $n \in \mathbb{N}$, we call S a finite set with cardinality n.

- Examples:

$A = \{\text{Mercedes}, \text{BMW}, \text{Porsche}\}$, $|A| = 3$

$B = \{1, \{2, 3\}, \{4, 5\}, 6\}$ $|B| = 4$

$C = \emptyset$ $|C| = 0$

$D = \{ x \in \mathbb{N} \mid x \leq 7000 \}$ $|D| = 7001$

$E = \{ x \in \mathbb{N} \mid x \geq 7000 \}$ E is infinite!

The Power Set

- $P(A)$ "power set of A" (also written as 2^A)
- $P(A) = \{B \mid B \subseteq A\}$ (contains all subsets of A)

- Examples:

- $A = \{x, y, z\}$
- $P(A) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$

- $A = \emptyset$
- $P(A) = \{\emptyset\}$
- Note: $|A| = 0$, $|P(A)| = 1$

Cartesian Product

- Example:
- A = {good, bad}, B = {student, prof}
- $A \times B$ = {(good, student), (good, prof), (bad, student), (bad, prof)}

$B \times A$ = { (student, good), (prof, good), (bad, student), (prof, bad)}

Example: A = {x, y}, B = {a, b, c}

$A \times B$ = {(x, a), (x, b), (x, c), (y, a), (y, b), (y, c)}

Cartesian Product

- Note that:
  - $A \times \varnothing = \varnothing$
  - $\varnothing \times A = \varnothing$
  - For non-empty sets A and B: $A \neq B \Leftrightarrow A \times B \neq B \times A$
  - $|A \times B| = |A| \cdot |B|$

- The Cartesian product of two or more sets is defined as:
- $A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } 1 \leq i \leq n\}$

Cartesian Product

- The ordered n-tuple $(a_1, a_2, a_3, \ldots, a_n)$ is an ordered collection of n objects.
- Two ordered n-tuples $(a_1, a_2, a_3, \ldots, a_n)$ and $(b_1, b_2, b_3, \ldots, b_n)$ are equal if and only if they contain exactly the same elements in the same order, i.e. $a_i = b_i$ for $1 \leq i \leq n$.

- The Cartesian product of two sets is defined as:
- $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$

Set Operations

- Union: $A \cup B = \{x \mid x \in A \vee x \in B\}$

- Example: A = {a, b}, B = {b, c, d}
- $A \cup B = \{a, b, c, d\}$

- Intersection: $A \cap B = \{x \mid x \in A \wedge x \in B\}$

- Example: A = {a, b}, B = {b, c, d}
- $A \cap B = \{b\}$
- Cardinality: $|A \cup B| = |A| + |B| - |A \cap B|$

Set Operations

- Two sets are called disjoint if their intersection is empty, that is, they share no elements:
- $A \cap B = \emptyset$

- The difference between two sets A and B contains exactly those elements of A that are not in B:
- $A-B = \{x \mid x \in A \wedge x \notin B\}$

Example: $A = \{a, b\}$, $B = \{b, c, d\}$, $A-B = \{a\}$

- Cardinality: $|A-B| = |A| - |A \cap B|$

Set Operations

- The complement of a set A contains exactly those elements under consideration that are not in A: denoted A^c (or as in the text)
- $A^c = U - A$

- Example: $U = \mathbf{N}$, $B = \{250, 251, 252, \ldots\}$
- $B^c = \{0, 1, 2, \ldots, 248, 249\}$

Logical Equivalence

- Equivalence laws
  - Identity laws, $P \wedge T \equiv P$,
  - Domination laws, $P \wedge F \equiv F$,
  - Idempotent laws, $P \wedge P \equiv P$,
  - Double negation law, $\neg (\neg P) \equiv P$
  - Commutative laws, $P \wedge Q \equiv Q \wedge P$,
  - Associative laws, $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$,
  - Distributive laws, $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$,
  - De Morgan's laws, $\neg (P \wedge Q) \equiv (\neg P) \vee (\neg Q)$
  - Law with implication $P \rightarrow Q \equiv \neg P \vee Q$

Set Identity

- Identity laws, $A \cup \varnothing = A,\ A \cap U = A$
- Domination laws, $A \cup U = U,\ A \cap \varnothing = \varnothing$
- Idempotent laws, $A \cup A = A,\ A \cap A = A$
- Complementation law, $(A^c)^c = A$
- Commutative laws, $A \cup B = B \cup A,\ A \cap B = B \cap A$
- Associative laws, $A \cup (B \cup C) = (A \cup B) \cup C, \ldots$
- Distributive laws, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \ldots$
- De Morgan's laws, $(A \cup B)^c = A^c \cap B^c\ ,\ (A \cap B)^c = A^c \cup B^c$
- Absorption laws, $A \cup (A \cap B) = A,\ A \cap (A \cup B) = A$
- Complement laws, $A \cup A^c = U,\ A \cap A^c = \varnothing$

Set Identity

- How can we prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$?
- Method I: logical equivalent
- $x \in A \cup (B \cap C)$

$\Leftrightarrow x \in A \vee x \in (B \cap C)$

$\Leftrightarrow x \in A \vee (x \in B \wedge x \in C)$

$\Leftrightarrow (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)$ (distributive law)

$\Leftrightarrow x \in (A \cup B) \wedge x \in (A \cup C)$

$\Leftrightarrow x \in (A \cup B) \cap (A \cup C)$

Every logical expression can be transformed into an equivalent expression in set theory and vice versa.

Question?

Functions

Lecture Five

Functions

- A function f from a set A to a set B is an assignment of exactly one element of B to each element of A.
- We write
- f(a) = b
- if b is the unique element of B assigned by the function f to the element a of A.

- If f is a function from A to B, we write
- f: A→B
- (note: Here, "→" has nothing to do with if… then)

Functions

- If $f:A \rightarrow B$, we say that A is the domain of f and B is the codomain of f.

- If $f(a) = b$, we say that b is the image of a and a is the pre-image of b.

- The range of $f:A \rightarrow B$ is the set of all images of all elements of A.

- We say that $f:A \rightarrow B$ maps A to B.

Functions

- Let us take a look at the function $f:P \rightarrow C$ with
- P = {Linda, Max, Kathy, Peter}
- C = {Boston, New York, Hong Kong, Moscow}

- f(Linda) = Moscow
- f(Max) = Boston
- f(Kathy) = Hong Kong
- f(Peter) = New York

- Here, the range of f is C.

Functions

- Let us re-specify f as follows:

- f(Linda) = Moscow
- f(Max) = Boston
- f(Kathy) = Hong Kong
- f(Peter) = Boston

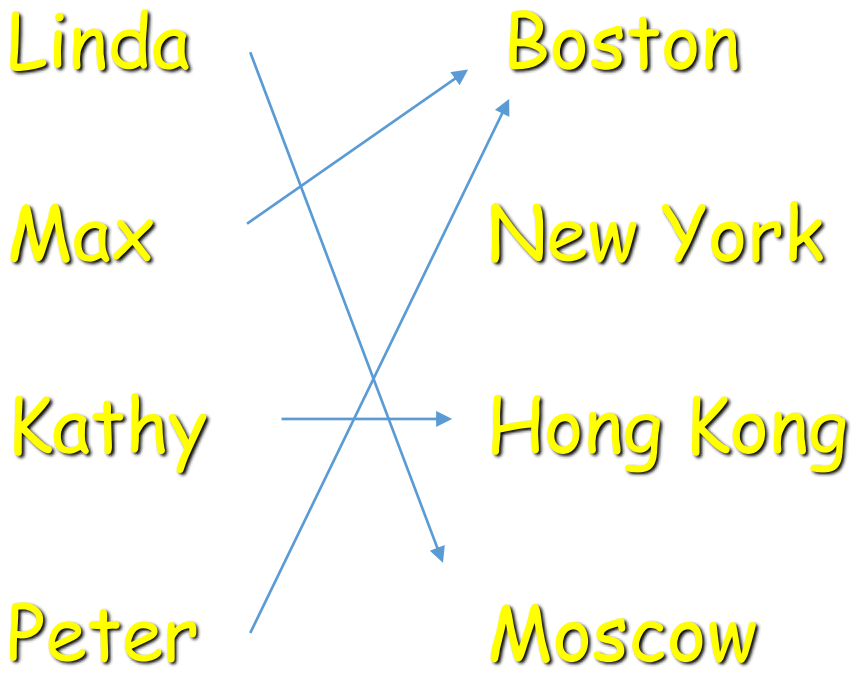
- Is f still a function?

What is its range? {Moscow, Boston, Hong Kong}

Functions

- Other ways to represent f:

x	f(x)
Linda	Moscow
Max	Boston
Kathy	Hong Kong
Peter	Boston



Functions

- If the domain of our function f is large, it is convenient to specify f with a formula, e.g.:

- $f: \mathbf{R} \rightarrow \mathbf{R}$
- $f(x) = 2x$

- This leads to:
- $f(1) = 2$
- $f(3) = 6$
- $f(-3) = -6$
- …

Functions

- Let f_1 and f_2 be functions from A to $\mathbf{R}$.
- Then the sum and the product of f_1 and f_2 are also functions from A to $\mathbf{R}$ defined by:
- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
- $(f_1 f_2)(x) = f_1(x)\, f_2(x)$

- Example:
- $f_1(x) = 3x$, $f_2(x) = x + 5$
- $(f_1 + f_2)(x) = f_1(x) + f_2(x) = 3x + x + 5 = 4x + 5$
- $(f_1 f_2)(x) = f_1(x)\, f_2(x) = 3x\,(x + 5) = 3x^2 + 15x$

Functions

- We already know that the range of a function $f: A \to B$ is the set of all images of elements $a \in A$.

- If we only regard a subset $S \subseteq A$, the set of all images of elements $s \in S$ is called the image of S.

- We denote the image of S by $f(S)$:

- $f(S) = \{f(s) \mid s \in S\}$

Functions

- Let us look at the following well-known function:
- f(Linda) = Moscow
- f(Max) = Boston
- f(Kathy) = Hong Kong
- f(Peter) = Boston

- What is the image of S = {Linda, Max} ?
- f(S) = {Moscow, Boston}

- What is the image of S = {Max, Peter} ?
- f(S) = {Boston}

Properties of Functions

- A function $f:A \to B$ is said to be one-to-one (or injective), if and only if

- $\forall x, y \in A\ (f(x) = f(y) \rightarrow x = y)$

- In other words: f is one-to-one if and only if it does not map two distinct elements of A onto the same element of B.

Properties of Functions

- And again…
- f(Linda) = Moscow
- f(Max) = Boston
- f(Kathy) = Hong Kong
- f(Peter) = Boston

- Is f one-to-one?

- No, Max and Peter are mapped onto the same element of the image.

g(Linda) = Moscow
g(Max) = Boston
g(Kathy) = Hong Kong
g(Peter) = New York

Is g one-to-one?

Yes, each element is assigned a unique element of the image.

Properties of Functions

- How can we prove that a function f is one-to-one?
- Whenever you want to prove something, first take a look at the relevant definition(s):
- $\forall x, y \in A\ (f(x) = f(y) \rightarrow x = y)$
- Example:
- $f: \mathbf{R} \rightarrow \mathbf{R}$
- $f(x) = x^2$
- Disproof by counterexample:
- $f(3) = f(-3)$, but $3 \neq -3$, so f is not one-to-one.

Properties of Functions

- … and yet another example:

- $f:\mathbf{R}\rightarrow\mathbf{R}$
- $f(x) = 3x$

- One-to-one: $\forall x, y \in A\ (f(x) = f(y) \rightarrow x = y)$
- To show: $f(x) \neq f(y)$ whenever $x \neq y$ (indirect proof)

- $x \neq y$

$\Leftrightarrow 3x \neq 3y$

$\Leftrightarrow f(x) \neq f(y)$,

so if $x \neq y$, then $f(x) \neq f(y)$, that is, f is one-to-one.

Properties of Functions

- A function $f:A \to B$ with $A,B \subseteq R$ is called strictly increasing, if
- $\forall x,y \in A\ (x < y \to f(x) < f(y))$,
- and strictly decreasing, if
- $\forall x,y \in A\ (x < y \to f(x) > f(y))$.

- Obviously, a function that is either strictly increasing or strictly decreasing is one-to-one.

Properties of Functions

- A function $f: A \to B$ is called onto, or surjective, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

- In other words, f is onto if and only if its range is its entire codomain.

- A function $f: A \to B$ is a one-to-one correspondence, or a bijection, if and only if it is both one-to-one and onto.

- Obviously, if f is a bijection and A and B are finite sets, then $|A| = |B|$.

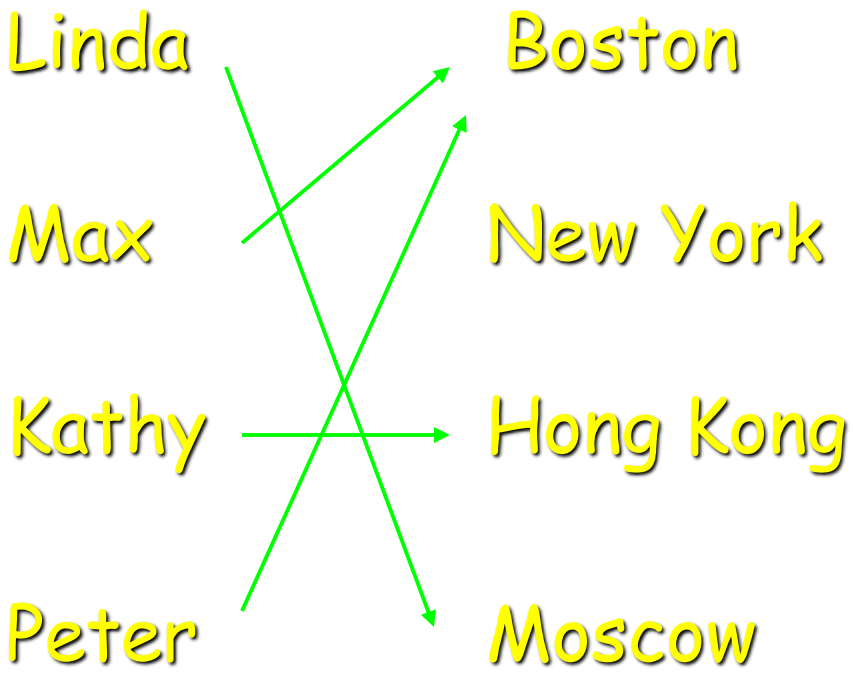
Properties of Functions

- Examples:

- In the following examples, we use the arrow representation to illustrate functions $f: A \rightarrow B$.

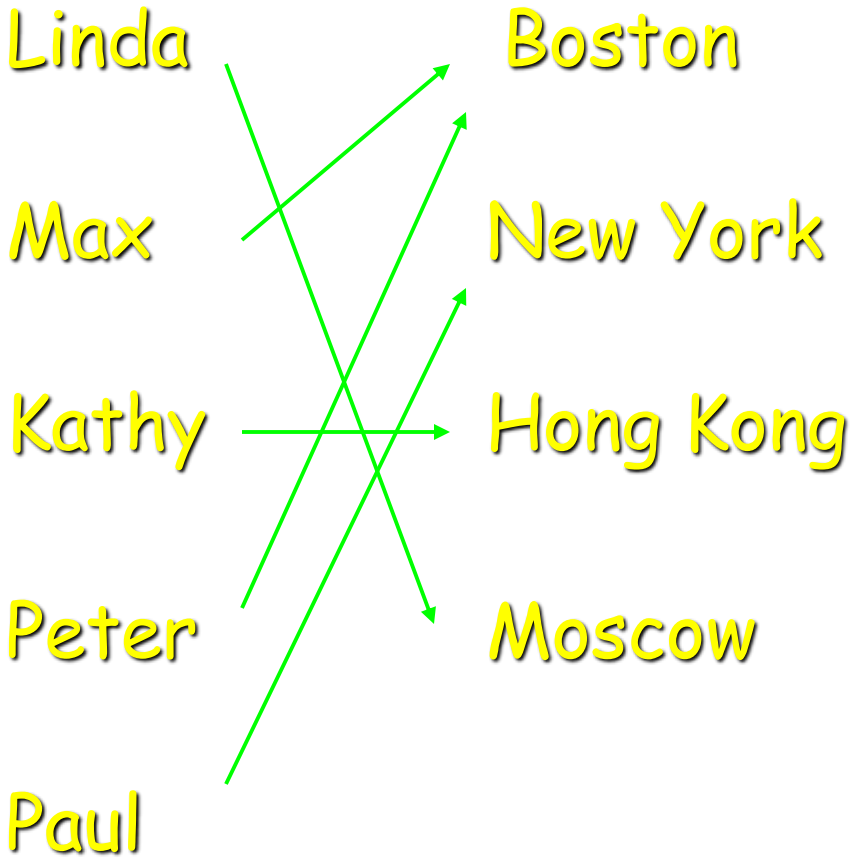
- In each example, the complete sets A and B are shown.

Properties of Functions



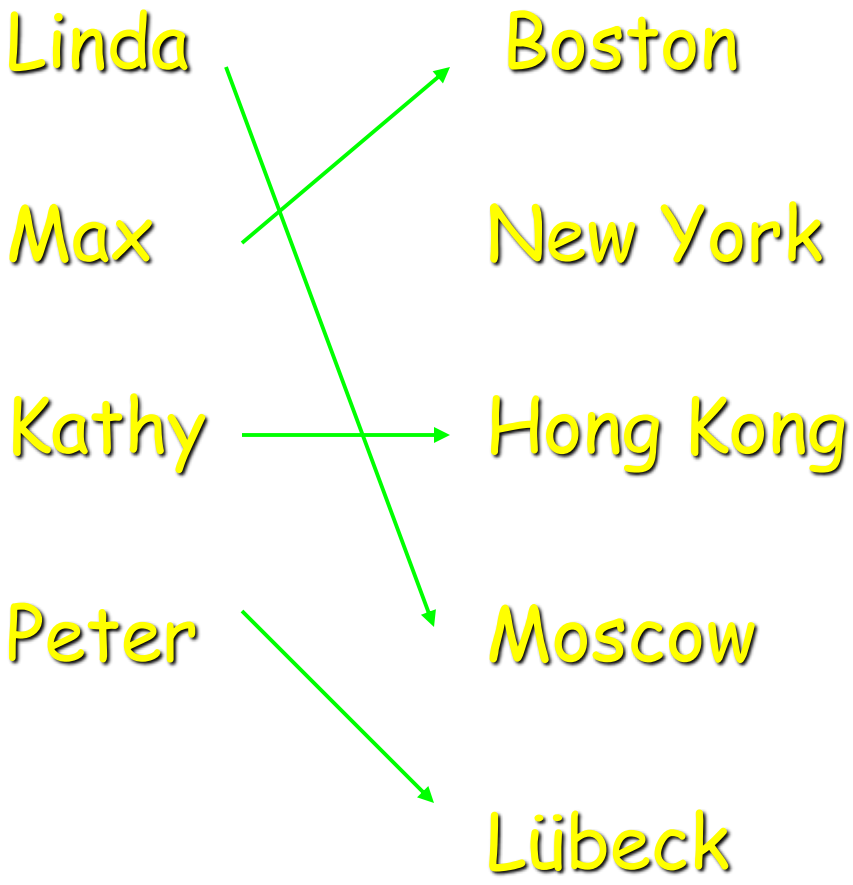
- Is f injective?
- No.
- Is f surjective?
- No.
- Is f bijective?
- No.

Properties of Functions



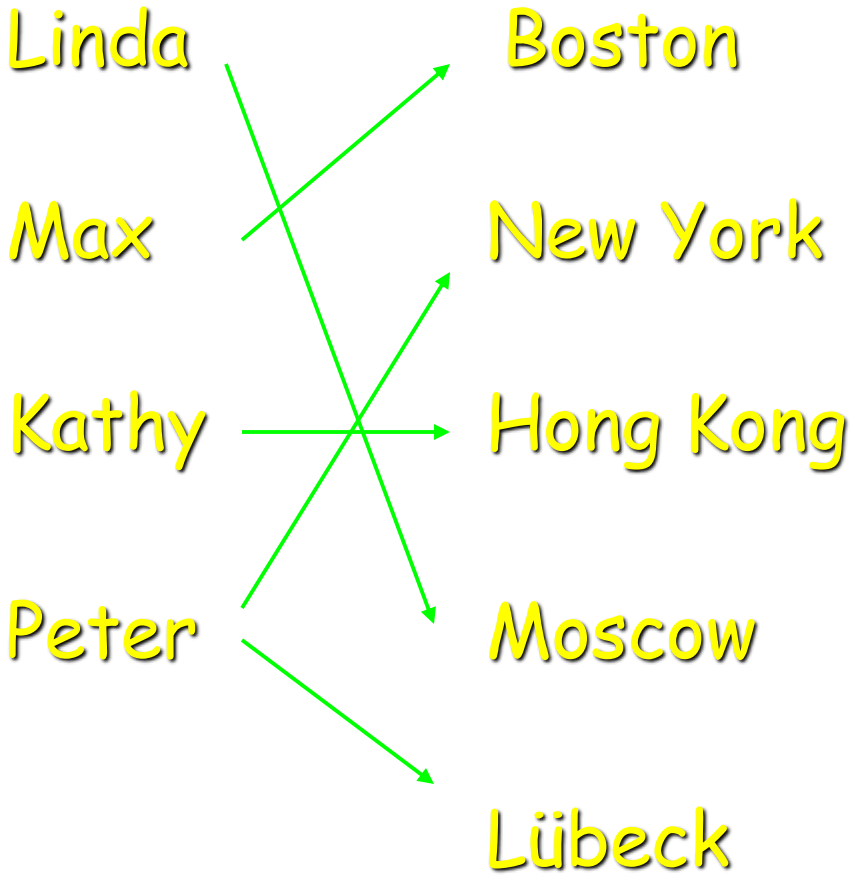
- Is f injective?
- No.
- Is f surjective?
- Yes.
- Is f bijective?
- No.

Properties of Functions



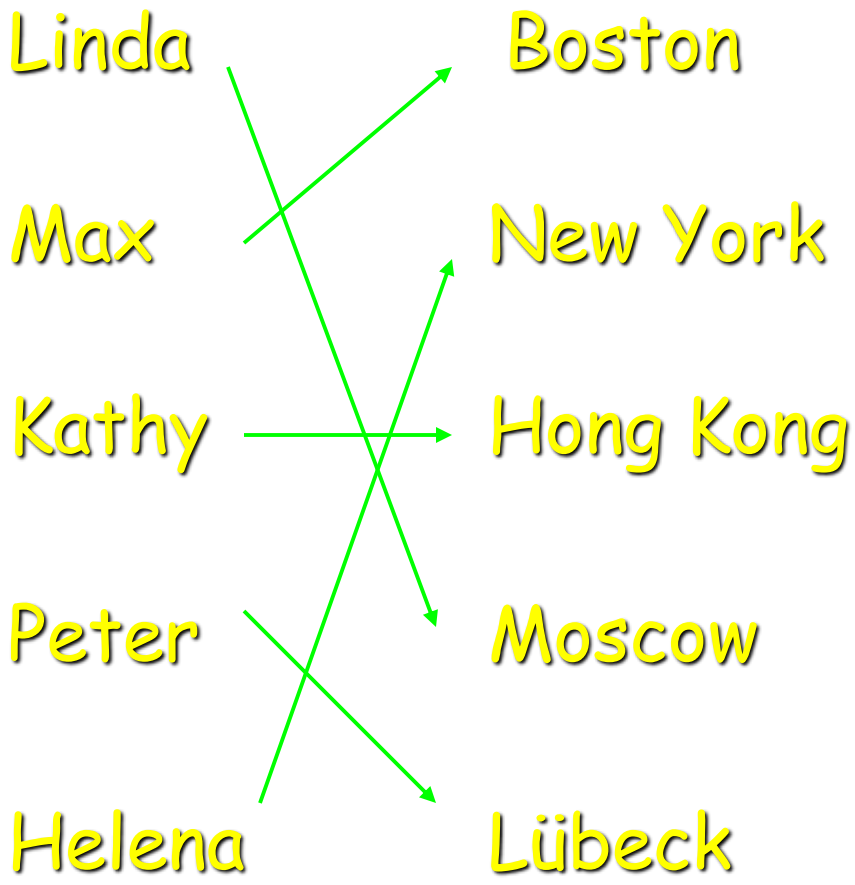
- Is f injective?
- Yes.
- Is f surjective?
- No.
- Is f bijective?
- No.

Properties of Functions



- Is f injective?
- No! f is not even a function!

Properties of Functions



- Is f injective?
- Yes.
- Is f surjective?
- Yes.
- Is f bijective?
- Yes.

Inversion

- An interesting property of bijections is that they have an **inverse function**.

- The **inverse function** of the bijection $f:A \to B$ is the function $f^{-1}:B \to A$ with
- $f^{-1}(b) = a$ whenever $f(a) = b$.

Inversion

- Example:

- f(Linda) = Moscow
- f(Max) = Boston
- f(Kathy) = Hong Kong
- f(Peter) = Lübeck
- f(Helena) = New York

- Clearly, f is bijective.

The inverse function f^{-1} is given by:

f^{-1}(Moscow) = Linda

f^{-1}(Boston) = Max

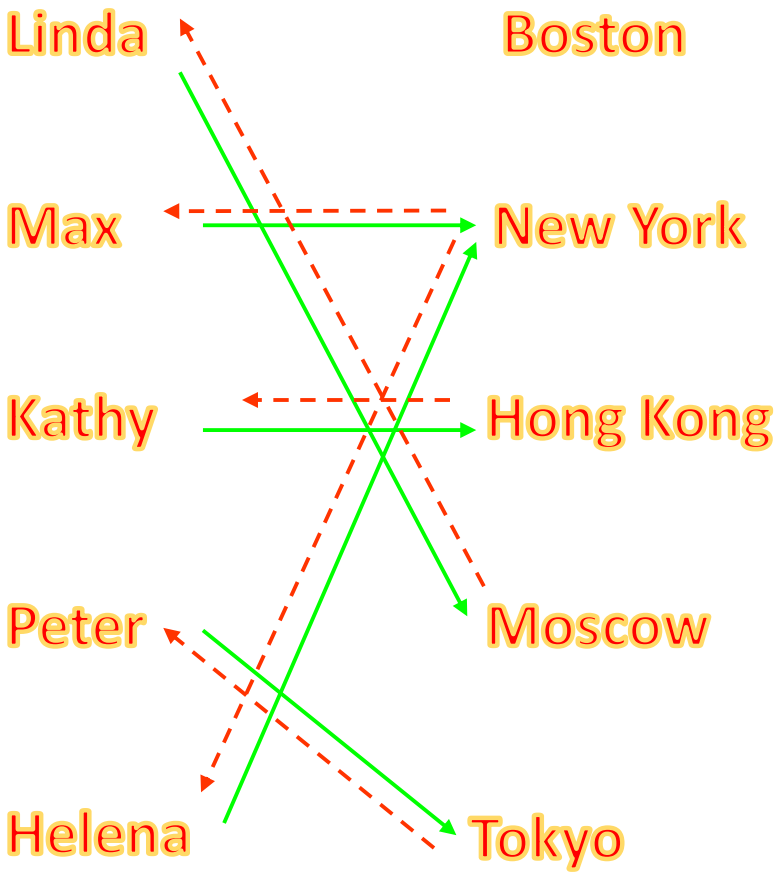
f^{-1}(Hong Kong) = Kathy

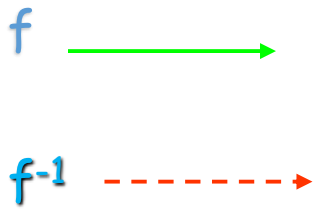
f^{-1}(Lübeck) = Peter

f^{-1}(New York) = Helena

Inversion is only possible for bijections (= invertible functions)

Inversion





- $f^{-1}: C \rightarrow P$ is no function, because it is not defined for all elements of C and assigns two images to the pre-image New York.

Composition

- The **composition** of two functions $g:A \to B$ and $f:B \to C$, denoted by $f \circ g$, is defined by

- $(f \circ g)(a) = f(g(a))$

- This means that
  - **first**, function g is applied to element $a \in A$, mapping it onto an element of B,
  - **then**, function f is applied to this element of B, mapping it onto an element of C.
  - **Therefore**, the composite function maps from A to C.

Composition

- Example:

- $f(x) = 7x - 4$, $g(x) = 3x$,
- $f: \mathbf{R} \to \mathbf{R}$, $g: \mathbf{R} \to \mathbf{R}$

- $(f \circ g)(5) = f(g(5)) = f(15) = 105 - 4 = 101$

- $(f \circ g)(x) = f(g(x)) = f(3x) = 21x - 4$

Composition

- Composition of a function and its inverse:

- $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x$

- The composition of a function and its inverse is the **identity function** $i(x) = x$.

Floor and Ceiling Functions

- The **floor** and **ceiling** functions map the real numbers onto the integers ($\mathbf{R} \rightarrow \mathbf{Z}$).

- The **floor** function assigns to $r \in \mathbf{R}$ the largest $z \in \mathbf{Z}$ with $z \leq r$, denoted by $\lfloor r \rfloor$.

- **Examples:** $\lfloor 2.3 \rfloor = 2, \lfloor 2 \rfloor = 2, \lfloor 0.5 \rfloor = 0, \lfloor -3.5 \rfloor = -4$

- The **ceiling** function assigns to $r \in \mathbf{R}$ the smallest $z \in \mathbf{Z}$ with $z \geq r$, denoted by $\lceil r \rceil$.

- **Examples:** $\lceil 2.3 \rceil = 3, \lceil 2 \rceil = 2, \lceil 0.5 \rceil = 1, \lceil -3.5 \rceil = -3$

QUESTION ???

Boolean algebra

Lecture Six

Boolean Algebra

- Boolean algebra provides the operations and the rules for working with the set {0, 1}.
- These are the rules that underlie electronic circuits.
- We are going to focus on three operations:
  - Boolean complementation,
  - Boolean sum, and
  - Boolean product

Boolean Operations

- The complement is denoted by a bar (on the slides, we will use a minus sign). It is defined by
- $-0 = 1$ and $-1 = 0$.

- The Boolean sum, denoted by + or by OR, has the following values:
- $1 + 1 = 1, \quad 1 + 0 = 1, \quad 0 + 1 = 1, \quad 0 + 0 = 0$

- The Boolean product, denoted by $\cdot$ or by AND, has the following values:
- $1 \cdot 1 = 1, \quad 1 \cdot 0 = 0, \quad 0 \cdot 1 = 0, \quad 0 \cdot 0 = 0$

Boolean Functions and Expressions

- Definition: Let B = {0, 1}. The variable x is called a Boolean variable if it assumes values only from B.
- A function from B^n, the set $\{(x_1, x_2, \ldots, x_n) \mid x_i \in B, 1 \leq i \leq n\}$, to B is called a Boolean function of degree n.
- Boolean functions can be represented using expressions made up from the variables and Boolean operations.

Boolean Functions and Expressions

- **Definition:** The Boolean functions F and G of n variables are **equal** if and only if $F(b_1, b_2, \ldots, b_n) = G(b_1, b_2, \ldots, b_n)$ whenever $b_1, b_2, \ldots, b_n$ belong to B.
- Two different Boolean expressions that represent the same function are called **equivalent**.
- For example, the Boolean expressions xy, xy + 0, and xy·1 are equivalent.

Boolean Functions and Expressions

- The **complement** of the Boolean function F is the function –F, where $-F(b_1, b_2, \ldots, b_n) = -(F(b_1, b_2, \ldots, b_n))$.
- Let F and G be Boolean functions of degree n. The **Boolean sum F+G** and **Boolean product FG** are then defined by
- $(F + G)(b_1, b_2, \ldots, b_n) = F(b_1, b_2, \ldots, b_n) + G(b_1, b_2, \ldots, b_n)$
- $(FG)(b_1, b_2, \ldots, b_n) = F(b_1, b_2, \ldots, b_n)\, G(b_1, b_2, \ldots, b_n)$

Boolean Functions and Expressions

- **Question:** How many different Boolean functions of degree 1 are there?
- **Solution:** There are four of them, F_1, F_2, F_3, and F_4:

x	F_1	F_2	F_3	F_4
0	0	0	1	1
1	0	1	0	1

Boolean Functions and Expressions

- **Question:** How many different Boolean functions of degree n are there?
- **Solution:**
- There are 2^n different n-tuples of 0s and 1s.
- A Boolean function is an assignment of 0 or 1 to each of these 2^n different n-tuples.
- Therefore, there are 2^{2^n} different Boolean functions.

Definition of a Boolean Algebra

- All the properties of Boolean functions and expressions that we have discovered also apply to other mathematical structures such as propositions and sets and the operations defined on them.
- If we can show that a particular structure is a Boolean algebra, then we know that all results established about Boolean algebras apply to this structure.
- For this purpose, we need an abstract definition of a Boolean algebra.

Definition of a Boolean Algebra

- Definition: A Boolean algebra is a set B with two binary operations $\vee$ and $\wedge$, elements 0 and 1, and a unary operation – such that the following properties hold for all x, y, and z in B:

➢ $x \vee 0 = x$ and $x \wedge 1 = x$ (identity laws)

➢ $x \vee (-x) = 1$ and $x \wedge (-x) = 0$ (domination laws)

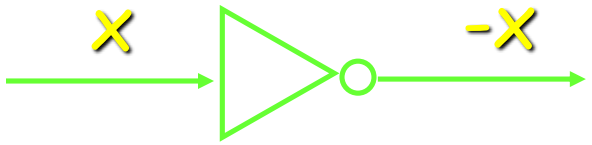
➢ $(x \vee y) \vee z = x \vee (y \vee z)$ and
$(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and (associative laws)

➢ $x \vee y = y \vee x$ and $x \wedge y = y \wedge x$ (commutative laws)

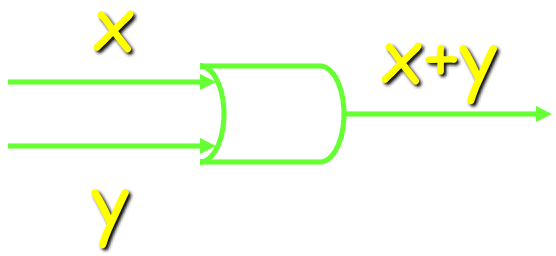
➢ $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ and
$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ (distributive laws)

Logic Gates

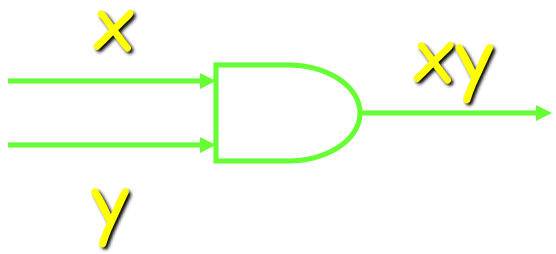
•Electronic circuits consist of so-called gates. There are three basic types of gates:



inverter



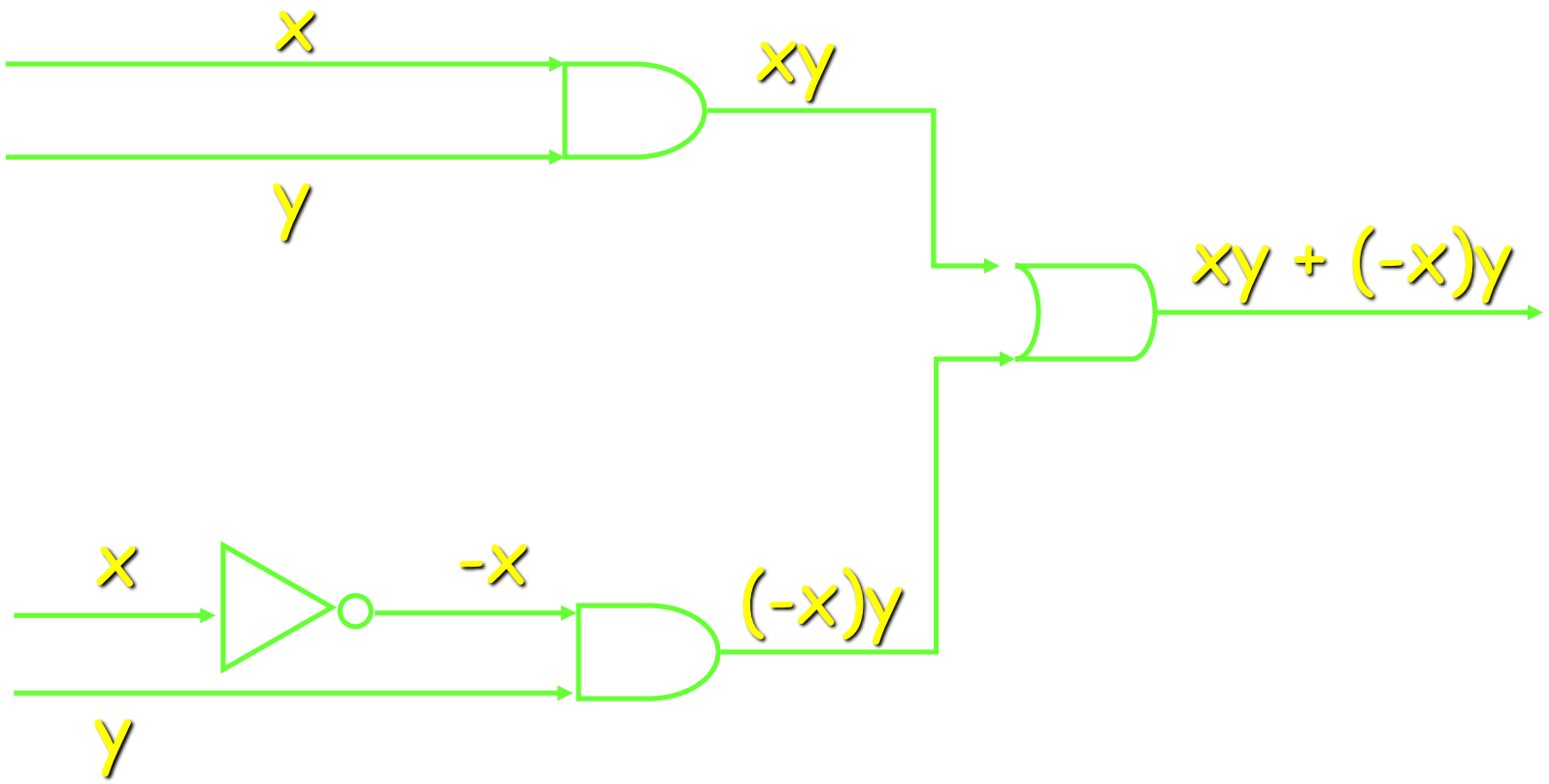
OR gate



AND gate

Logic Gates

- **Example:** How can we build a circuit that computes the function xy + (-x)y ?



Logic, Sets, and Boolean Algebra

Logic	Set	Boolean Algebra
False	$\varnothing$	0
True	U	1
$A \wedge B$	$A \cap B$	$A \cdot B$
$A \vee B$	$A \cup B$	$A+B$
$\neg A$	A^c	

Compare the equivalence laws of them

END…
QUESTION???

Sequences

Lecture 7

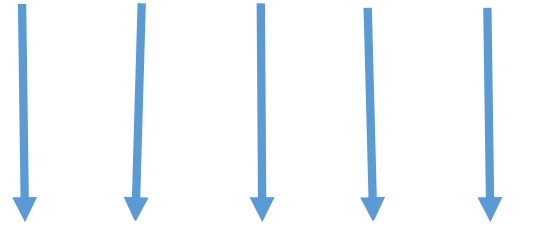
Sequences

Sequences represent ordered lists of elements.

A sequence is defined as a function from a subset of N to a set S. We use the notation an to denote the image of the integer n. We call an a term of the sequence.

Example:

subset of N: 1 2 3 4 5 …



- S= 2 4 6 8 10

Sequences

We use the notation $\{a_n\}$ to describe a sequence.

Important: Do not confuse this with the {} used in set notation.

It is convenient to describe a sequence with a formula.

For example, the sequence on the previous slide can be specified as $\{a_n\}$, where $a_n = 2n$.

The Formula Game

What are the formulas that describe the following sequences a1, a2, a3, … ?

1, 3, 5, 7, 9, … → $a_n = 2n - 1$

1, 1, -1, 1, -1, … → $a_n = (-1)^n$

2, 5, 10, 17, 26, … → $a_n = n^2 + 1$

0.25, 0.5, 0.75, 1, 1.25 … → $a_n = 0.25n$

3, 9, 27, 81, 243, … → $a_n = 3^n$

Strings

Finite sequences are also called strings, denoted by $a_1a_2a_3...a_n$.

The length of a string S is the number of terms that it consists of.

The empty string contains no terms at all. It has length zero.

Summations

What does $\sum_{j=m}^{n} a_j$ stand for?

It represents the sum $a_m + a_{m+1} + a_{m+2} + \ldots + a_n$.

The variable j is called the index of summation, running from its lower limit m to its upper limit n. We could as well have used any other letter to denote this index.

Summations

How can we express the sum of the first 1000 terms of the sequence $\{a_n\}$ with $a_n=n^2$ for

$n = 1, 2, 3, \ldots$?

We write it as $\sum_{j=1}^{1000} j^2$

- What is the value of $\sum_{j=1}^{6} j$? It is 1 + 2 + 3 + 4 + 5 + 6 = 21.

It is so much work to calculate this…

Summations

It is said that Friedrich Gauss came up with the following formula:

$$\sum_{j=1}^{n} n = \frac{n(n+1)}{2}$$

When you have such a formula, the result of any summation can be calculated much more easily, for example:

$$\sum_{j=1}^{100} j = \frac{100(100+1)}{2} = \frac{10100}{2} = 5050$$

Arithmetic Series

- How does: $\sum_{j=1}^{n} j = \frac{n(n+1)}{2}$???
- Observe that:

1 + 2 + 3 +…+ n/2 + (n/2 + 1) +…+ (n - 2) + (n - 1) + n

= [1 + n] + [2 + (n - 1)] + [3 + (n - 2)] +…+ [n/2 + (n/2 + 1)]

= (n + 1) + (n + 1) + (n + 1) + … + (n + 1) (with n/2 terms)

= n(n + 1)/2.

Geometric Series

How does: $\sum_{j=0}^{n} a^j = \dfrac{a^{n+1}-1}{(a-1)}$

Observe that:

$S = 1 + a + a^2 + a^3 + \ldots + a^n$

$a*S = a + a^2 + a^3 + \ldots + a^n + a^{(n+1)}$

so, $(aS - S) = (a - 1)\,S = a^{(n+1)} - 1$

Therefore, $1 + a + a^2 + \ldots + a^n = (a^{(n+1)} - 1) / (a - 1)$.

For example: 1 + 2 + 4 + 8 +... + 1024 = 2047.

Useful Series

- $\sum_{j=1}^{n} j = \frac{n(n+1)}{2}$
- $\sum_{j=0}^{n} a^j = \frac{a^{(n+1)} - 1}{(a-1)}$
- $\sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{j=1}^{n} j^3 = \frac{n^2 (n+1)^2}{4}$

Double Summations

Corresponding to nested loops in C or Java, there is also double (or triple etc.) summation:

Example:

$$\sum_{i=1}^{5}\sum_{j=1}^{2} ij$$

$$=\sum_{i=1}^{5}(i+2i)$$

$$= \sum_{i=1}^{5} 3i \qquad = 3+6+9+12+15 = 45$$

Questions



Mathematical Induction

Lecture 8

Mathematical Induction

The principle of Mathematical Induction is a useful tool for proving that a certain predicate is true for all natural numbers.

- It cannot be used to discover theorems, but only to prove them.

Induction

If we have a propositional function P(n), and we want to prove that P(n) is true for any natural number n, we do the following:

Show that P(0) is true.
(basic step)

Show that if P(n) then P(n + 1) for any $n \in \mathbb{N}$.
(inductive step)

Then P(n) must be true for any $n \in \mathbb{N}$.
(conclusion)

Induction

Example:

Show that $n < 2^n$ for all positive integers n.

Let P(n) be the proposition "$n < 2^n$."

1. Show that P(1) is true.
(basis step)

P(1) is true, because $1 < 2^1 = 2$.

Induction

2. Show that if P(n) is true, then P(n + 1) is true.
(inductive step)

Assume that $n < 2^n$ is true.

We need to show that P(n + 1) is true, i.e.

$n + 1 < 2^{n+1}$

We start from $n < 2^n$:

$n + 1 < 2^n + 1 \leq 2^n + 2^n = 2^{n+1}$

Therefore, if $n < 2^n$ then $n + 1 < 2^{n+1}$

Induction

Then P(n) must be true for any positive integer. (conclusion)

$n < 2^n$ is true for any positive integer.

End of proof.

Induction

Another Example ("Gauss"):

$1 + 2 + \ldots + n = n\,(n + 1)/2$

Show that P(0) is true.
(basis step)

For n = 0 we get 0 = 0. True.

Induction

Show that if $P(n)$ then $P(n + 1)$ for any $n \in \mathbb{N}$. (inductive step)

$$1 + 2 + \ldots + n = n(n + 1)/2$$

$$1 + 2 + \ldots + n + (n + 1) = n(n + 1)/2 + (n + 1)$$

$$= (2n + 2 + n(n + 1))/2$$

$$= (2n + 2 + n^2 + n)/2$$

$$= (2 + 3n + n^2)/2$$

$$= (n + 1)(n + 2)/2$$

$$= (n + 1)((n + 1) + 1)/2$$

Induction

Then $P(n)$ must be true for any $n \in \mathbb{N}$. (conclusion)

$1 + 2 + \ldots + n = n\,(n + 1)/2$ is true for all $n \in \mathbb{N}$.

End of proof.

Working With Quantifiers

Tutorial 2

1.Universal Quantifiers. "$\forall\ x$"

Problem...1

Let P(x) be the statement "x + 1 > x". What is the truth value of the quantification ∀x P(x), where the domain consists of all real numbers?

Solution: Because P(x) is true for all real numbers x, the quantification: ∀x P(x) is true.

Problem...2

Let Q(x) be the statement "x < 2". What is the truth value of the quantification ∀x Q(x), where the domain consists of all real numbers?

Solution: Q(x) is not true for every real number x, because, for instance, Q(3) is false. That is,

x = 3 is a counterexample for the statement ∀x Q(x). Thus

∀x Q(x) is false.

Problem...3

Suppose that P(x) is "$x^2 > 0$" . What is the truth value of the quantification $\forall x\ P(x)$, where the universe of discourse consists of all integers.

- Solution. We see that $x = 0$ is a counterexample because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$. so P(x) is false.

Problem...4

What is the truth value of $\forall x\, P(x)$, where $P(x)$ is the statement "$x^2 < 10$" and the domain consists of the positive integers not exceeding 4?

Solution. The statement $\forall x\, P(x)$ is the same as the conjunction

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4),$$

Because $P(4)$, which is the statement "$4^2 < 10$," is false, it follows that $\forall x\, P(x)$ is false.

Problem...5

What does the statement $\forall x\, N(x)$ mean if $N(x)$ is "Computer x is connected to the network" and the domain consists of all computers on campus?

Solution: The statement $\forall x\, N(x)$ means that for every computer x on campus, that computer x is connected to the network. This statement can be expressed in English as "Every computer on campus is connected to the network".

2.The *Existential* Quantifier "$\exists x$"

Problem...6

Let $P(x)$ denote the statement "$x > 3$". What is the truth value of the quantification $\exists x\, P(x)$, where the domain consists of all real numbers?

Solution: Because "$x > 3$" is sometimes true—for instance, when $x = 4$—the existential quantification of $P(x)$, which is $\exists x\, P(x)$, is true.

Problem...7

Let $Q(x)$ denote the statement "$x = x + 1$" What is the truth value of the quantification $\exists x\ Q(x)$, where the domain consists of all real numbers?

Solution: Because $Q(x)$ is false for every real number x, the existential quantification of $Q(x)$, which is $\exists x\ Q(x)$, is false.

Problem...8

What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement "$x^2 > 10$" and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists x P(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4).$$

Because $P(4)$, which is the statement "$4^2 > 10$," is true, it follows that $\exists x P(x)$ is true.

Quantifiers.

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

3.Negation “¬”

Problem...9

What are the negations of the statements "There is an honest politician" and "All Americans eat cheeseburgers"?

Solution:

Part 1. Let $H(x)$ denote "x is honest". Then the statement "There is an honest politician" is represented by $\exists x H(x)$, where the domain consists of all politicians. The negation of this statement is $\neg\exists x H(x)$, which is equivalent to $\forall x \neg H(x)$. This negation can be expressed as "Every politician is dishonest."

Problem...9, cont.

Part 2. Let $C(x)$ denote "x eats cheeseburgers." Then the statement "All Americans eat cheeseburgers" is represented by $\forall x C(x)$, where the domain consists of all Americans. The negation of this statement is $\neg \forall x C(x)$, which is equivalent to $\exists x \neg C(x)$. This negation can be expressed in several different ways, including "Some American does not eat cheeseburgers" and "There is an American who does not eat cheeseburgers."

Problem...10

What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Solution: The negation of $\forall x(x^2 > x)$ is the statement $\neg\forall x(x^2 > x)$, which is equivalent to $\exists x\neg(x2 > x)$. This can be rewritten as $\exists x(x2 \leq x)$. The negation of $\exists x(x2 = 2)$ is the statement $\neg\exists x(x2 = 2)$, which is equivalent to $\forall x\neg(x2 = 2)$. This can be rewritten as $\forall x(x2 = 2)$.

The truth values of these statements depend on the domain

Problem...11

Show that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution: By De Morgan's law for universal quantifiers, we know that $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(\neg(P(x) \rightarrow Q(x)))$

we know that $\neg(P(x) \rightarrow Q(x)) \equiv P(x) \wedge \neg Q(x)$ for every x.

So, $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

De Morgan's Laws for Quantifiers.

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

Exercise

- Express the statement using predicates and quantifiers.

1. "Every student in this class has studied calculus".

2. "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico"

Exercise

- Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

1. "All hummingbirds are richly colored."
2. "No large birds live on honey."
3. "Birds that do not live on honey are dull in color."
4. "Hummingbirds are small."

Solution

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements respectively. Assuming that the domain consists of all birds,

Solution: We can express the statements in the argument as

1. $\forall x(P(x) \rightarrow S(x))$.
2. $\neg\exists x(Q(x) \wedge R(x))$.
3. $\forall x(\neg R(x) \rightarrow \neg S(x))$.
4. $\forall x(P(x) \rightarrow \neg Q(x))$.

Recursion

Lecture 9

Recursion

Recursion is a principle closely related to mathematical induction.

In a recursive definition, an object is defined in terms of itself.

- In recursion we nee to define

i. Initial value ii. The rule of construction

We can recursively define sequences, functions and sets.

Recursively Defined Sequences

Example:

The sequence $\{a_n\}$ of powers of 2 is given by

$a_n = 2^n$ for $n = 0, 1, 2, \ldots$.

The same sequence can also be defined recursively:

$a_0 = 1$ (initial value)

$a_{n+1} = 2a_n$ for $n = 0, 1, 2, \ldots$(the rule of sequence construction)

Obviously, induction and recursion are similar principles.

Recursively Defined Functions

We can use the following method to define a function with the natural numbers as its domain:

- Base case: Specify the value of the function at zero (initial value).
- Recursion: Give a rule for finding its value at any integer from its values at smaller integers (the of elements construction).

- Such a definition is called recursive or inductive definition.

Recursively Defined Functions

Example:

f(0) = 3 (initial value)
f(n + 1) = 2f(n) + 3 (the rule of element construction)

f(0) = 3
f(1) = 2*f(0) + 3 = 2*3 + 3 = 9
f(2) = 2*f(1) + 3 = 2*9 + 3 = 21
f(3) = 2*f(2) + 3 = 2*21 + 3 = 45
f(4) = 2*f(3) + 3 = 2*45 + 3 = 93

Recursively Defined Functions

How can we recursively define the factorial function $f(n) = n!$?

$f(0) = 1$

$f(n + 1) = (n + 1)*f(n)$

$f(0) = 1$

$f(1) = 1*f(0) = 1*1 = 1$

$f(2) = 2*f(1) = 2*1 = 2$

$f(3) = 3*f(2) = 3*2 = 6$

$f(4) = 4*f(3) = 4*6 = 24$

Recursively Defined Functions

A famous example: The Fibonacci numbers

$f(0) = 0$, $f(1) = 1$ (initial value)

$f(n) = f(n-1) + f(n-2)$ (the rule of construction)

So,

$f(0) = 0$

$f(1) = 1$

$f(2) = f(1) + f(0) = 1 + 0 = 1$

$f(3) = f(2) + f(1) = 1 + 1 = 2$

$f(4) = f(3) + f(2) = 2 + 1 = 3$

$f(5) = f(4) + f(3) = 3 + 2 = 5$

$f(6) = f(5) + f(4) = 5 + 3 = 8$

Recursively Defined Sets

If we want to recursively define a set, we need to provide two things:

- an initial set of elements,
- rules for the construction of additional elements from elements in the set.

Example: Let **S** be recursively defined by:

$3 \in S$

$(x + y) \in S$ if $(x \in S)$ and $(y \in S)$

S is the set of positive integers divisible by 3.

Recursively Defined Sets

Proof:

Let A be the set of all positive integers divisible by 3.

To show that $A = S$, we must show that

$A \subseteq S$ and $S \subseteq A$.

Part I: To prove that $A \subseteq S$, we must show that every positive integer divisible by 3 is in S.

We will use mathematical induction to show this.

Recursively Defined Sets

Let P(n) be the statement "3n belongs to S".

Basis step: P(1) is true, because 3 is in S.

Inductive step: To show:
If P(n) is true, then P(n + 1) is true.

Assume 3n is in S. Since 3n is in S and 3 is in S, it follows from the recursive definition of S that
3n + 3 = 3(n + 1) is also in S.

Conclusion of Part I: $A \subseteq S$.

Recursively Defined Sets

Part II: To show: $S \subseteq A$.

Basis step: To show:
All initial elements of S are in A. 3 is in A. True.

Inductive step: To show:
If x and y in S are in A, then (x + y) is in A .

Since x and y are both in A, it follows that $3 \mid x$ and $3 \mid y$. From Theorem I, Section 2.3, it follows that $3 \mid (x + y)$.

Conclusion of Part II: $S \subseteq A$.
Overall conclusion: $A = S$.

Questions

